

Day 169/365

Logistic Regression: predicting yes or no

A simple guide with everyday examples.

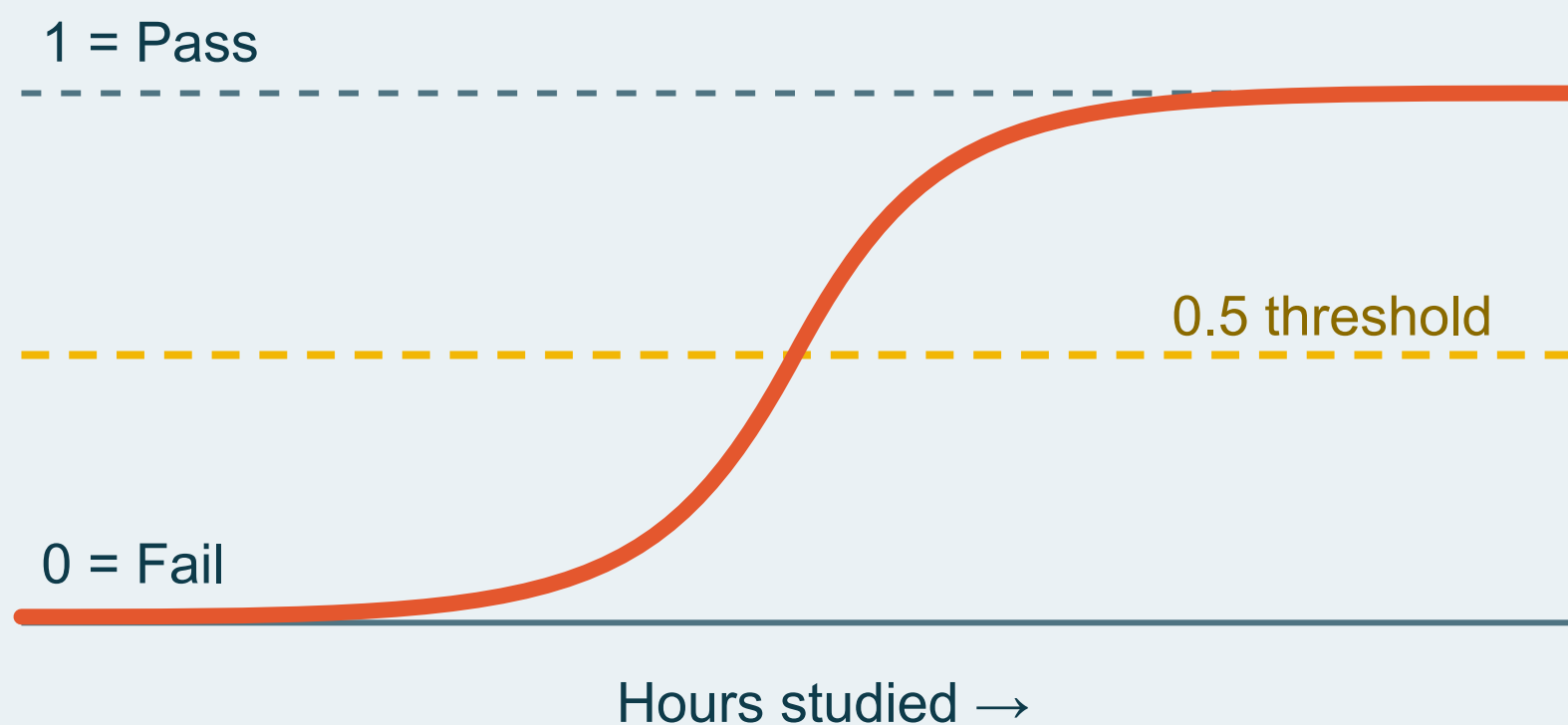
Numbers vs. categories

My car price series predicted a **number**. But many real questions have a category as the answer:

Question	Answer
Is this email spam?	Yes / No
Will the student pass?	Pass / Fail
Will the customer buy?	Buy / Don't buy
Is this tumour malignant?	Malignant / Benign

Despite the name, logistic regression is a **classification** model. It gives the **probability** of belonging to a class.

Why not just use a straight line?



A straight line can output 1.4 or -0.3 , which makes no sense as a probability.

The **sigmoid** curve squeezes any number into the 0 to 1 range. Far left is close to 0, far right is close to 1.

How it works in 3 steps

1. Score: combine the features, just like linear regression (weights $\times$ features + a starting value).

2. Sigmoid: turn that score into a probability.

3. Threshold: if the probability is 0.5 or more, predict class 1. Otherwise, class 0.

Student	Score	Probability of passing	Result
Asha	+0.8	0.69	Pass
Ravi	-1.5	0.18	Fail

Illustrative numbers.

Reading the coefficients: odds

Odds compare "happens" with "doesn't happen". A 75% chance means 75 : 25, or 3 to 1.

The model treats the log of the odds as a straight-line formula. That's why the coefficients are a little different from linear regression.

Example: the coefficient for "hours studied" is 0.7. Each extra hour multiplies the odds of passing by $e^{0.7} \approx 2.0$. So the odds roughly double.

Same trick as my Day 158 post, where I converted log-price coefficients with $\exp(\text{coef}) - 1$. Coefficients are in log space, so convert them before explaining.

Illustrative coefficient.

Three types

Type	Categories	Example
Binomial	Two	Spam or not spam
Multinomial	3+, no order	Cat, dog or sheep
Ordinal	3+, with order	Low, medium or high risk

For multinomial, the sigmoid is swapped for **softmax**, which gives one probability per class, all adding up to 1.

Example: an image of a handwritten digit gets 10 probabilities, one for each of 0 to 9. The highest wins.

Accuracy alone can fool you

Example: 100 patients, only 5 are sick. A model that says "healthy" for everyone is 95% accurate, yet it catches zero sick patients.

Metric	The question it answers
Precision	Of those I flagged, how many were right?
Recall	Of all real positives, how many did I catch?
F1	One score balancing the two

For the sick-patient case, recall matters most. For spam, precision matters more, since you don't want real mail in the spam folder.

3 things to remember

- 1. It predicts probabilities, then a threshold makes the call.**
- 2. Coefficients live in log-odds, so convert them.**
- 3. Judge it with precision and recall, not accuracy alone.**

It works best with independent rows, no extreme outliers and enough data.

Where would you use a yes/no prediction:
spam, churn or loans?